

DHCP Lab

dhcp-ethereal-trace-1.pcap

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bootp

No.	Time	Source	Destination	Protocol	Length	Info
2	7.587185	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x3e5e0ce3
4	8.632950	192.168.1.1	255.255.255.255	DHCP	590	DHCP Offer - Transaction ID 0x3e5e0ce3
5	8.633123	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0x3e5e0ce3
6	8.635133	192.168.1.1	255.255.255.255	DHCP	590	DHCP ACK - Transaction ID 0x3e5e0ce3
36	20.134178	192.168.1.101	192.168.1.1	DHCP	342	DHCP Request - Transaction ID 0x257e55a3
37	20.135930	192.168.1.1	255.255.255.255	DHCP	590	DHCP ACK - Transaction ID 0x257e55a3
41	25.073867	192.168.1.101	192.168.1.1	DHCP	342	DHCP Release - Transaction ID 0xb7a32733
42	30.869153	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x3a5df7d9
44	31.908133	192.168.1.1	255.255.255.255	DHCP	590	DHCP Offer - Transaction ID 0x3a5df7d9
45	31.908304	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0x3a5df7d9
46	31.910313	192.168.1.1	255.255.255.255	DHCP	590	DHCP ACK - Transaction ID 0x3a5df7d9

Frame 2: Packet, 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface 0

Ethernet II, Src: Dell_4f:36:23 (00:08:74:4f:36:23), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255

User Datagram Protocol, Src Port: 68, Dst Port: 67

Source Port: 68
Destination Port: 67
Length: 308
Checksum: 0xe97b [unverified]
[Checksum Status: Unverified]
[Stream index: 1]
[Stream Packet Number: 1]
[Timestamps]

0000 ff ff ff ff ff ff 00 08 74 4f 36 23 08 00 45 00t06# E-
0010 01 48 b3 10 00 00 11 86 95 00 00 00 00 ff ffH
0020 ff ff 00 44 00 43 01 34 e9 7b 01 01 06 00 3e 5eD C 4 {>
0030 0c e3 00 00 00 00 00 00 00 00 00 00 00 00 00 00t06#
0040 00 00 00 00 00 00 00 08 74 4f 36 23 00 00 00 00
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0090 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

dhcp-ethereal-trace-1.pcap

Packets: 63 - Displayed: 11 (17.5%)

Profile: Default

1. DHCP messages are sent over UDP. The client sends from port 68 and the server listens on port 67.

2. Client Server
|----- DHCP Discover ----->| (src:68, dst:67)
|<----- DHCP Offer -----| (src:67, dst:68)
|----- DHCP Request ----->| (src:68, dst:67)
|<----- DHCP ACK -----| (src:67, dst:68)

3. Source: Dell_4f:36:23 (00:08:74:4f:36:23)

4. Key differences:

Field	DHCP Discover	DHCP Request
Message Type	option DHCP Discover	DHCP Request
Requested IP Address	0.0.0.0 (none yet)	192.168.2.145 (the offered IP)
Server Identifier	Not present	Present (identifies which server it chose)

The Discover is essentially the client saying "I need an IP, anyone there?", it doesn't know any server yet. The Request is the client saying "I'd like the IP that server X offered me", it now has a specific server and address in mind.

5. DHCP Transaction ID Tracking
Packet Transaction ID
DHCP Discover 0xe220d8c
DHCP Offer 0xe220d8c

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DHCP Request 0xe220d8c
DHCP ACK 0xe220d8c

Observation:

All four packets share the same Transaction ID. This is intentional and indicates they belong to the same DHCP exchange.

Second Exchange (Request / ACK)

Packet Transaction ID

DHCP Request 0xb40714e
DHCP ACK 0xb40714e

Observation:

Both packets share the same Transaction ID, but it is different from the first exchange.

Purpose of Transaction ID:

The Transaction ID is a random number generated by the client at the start of a DHCP exchange. It acts like a tracking number that links all related messages together (Discover, Offer, Request, ACK). This allows the client to match server responses to its own request and keep multiple DHCP exchanges separate.

6. DHCP Packet IP Addressing

	Packet Source IP	Destination IP
DHCP Discover	0.0.0.0	255.255.255.255
DHCP Offer	192.168.2.1	255.255.255.255
DHCP Request	0.0.0.0	255.255.255.255
DHCP ACK	192.168.2.1	255.255.255.255

Why these addresses?

Client side (Discover & Request):

The source is 0.0.0.0 because the client does not yet have an IP address — this is the whole reason DHCP is being used.

The destination is 255.255.255.255 (broadcast) because the client does not know the DHCP server's address, so it sends the message to everyone on the local network.

Server side (Offer & ACK):

The source is 192.168.2.1, which is the DHCP server's IP address.

The destination remains 255.255.255.255 because the client still does not have a confirmed IP address at this stage, so the response is broadcast.

Therefore, The actual IP address being offered to the client is not in the IP header.

Instead, it is carried inside the DHCP message payload. That is why the IP header still uses 0.0.0.0 and broadcast addresses during most of the exchange.

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bootp

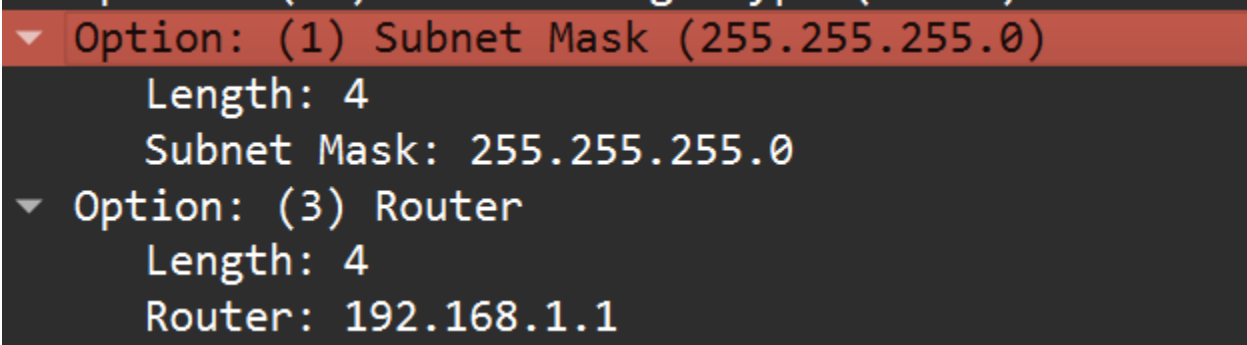
No.	Time	Source	Destination	Protocol	Length	Info
2	7.587185	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x3e5e0ce3
4	8.632950	192.168.1.1	255.255.255.255	DHCP	590	DHCP Offer - Transaction ID 0x3e5e0ce3
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46	31.910313	192.168.1.1	255.255.255.255	DHCP	590	DHCP ACK - Transaction ID 0x3a5df7d9

Hardware address length: 6
Hops: 0
Transaction ID: 0x3e5e0ce3
Seconds elapsed: 0
Bootp flags: 0x0000 (Unicast)
Client IP address: 0.0.0.0
Your (client) IP address: 192.168.1.101
Next server IP address: 0.0.0.0
Relay agent IP address: 0.0.0.0
Client MAC address: Dell_4f:36:23 (00:08:74:4f:36:23)
Client hardware address padding: 00000000000000000000
Server host name not given

0030 0c e3 00 00 00 00 00 00 00 c0 a8 01 65 00 00t06#....
0040 00 00 00 00 00 00 00 74 4f 36 23 00 00 00
0050 00 00 00 00 00 00 00 00 00 00 00 00 00
0060 00 00 00 00 00 00 00 00 00 00 00 00 00
0070 00 00 00 00 00 00 00 00 00 00 00 00 00
0080 00 00 00 00 00 00 00 00 00 00 00 00 00
0090 00 00 00 00 00 00 00 00 00 00 00 00 00
00a0 00 00 00 00 00 00 00 00 00 00 00 00 00
00b0 00 00 00 00 00 00 00 00 00 00 00 00 00
00c0 00 00 00 00 00 00 00 00 00 00 00 00 00
00d0 00 00 00 00 00 00 00 00 00 00 00 00 00
00e0 00 00 00 00 00 00 00 00 00 00 00 00 00
00f0 00 00 00 00 00 00 00 00 00 00 00 00 00
0100 00 00 00 00 00 00 00 00 00 00 00 00 00

Packets: 63 : Displayed: 11 (17.5%) Profile: Default

7. The DHCP server's IP address is **192.168.2.1**
8. Your (client) IP address: **192.168.1.101**
9. Relay agent IP address: 0.0.0.0

10. 

Subnet Mask:

Tells the client which part of the IP address identifies the network and which part identifies the host. For example, 255.255.255.0 means the first 3 octets (192.168.1) = network, last octet = host. The client uses this to determine whether a destination is on the same local network or needs to be routed elsewhere

Router (Default Gateway):

Tells the client the IP address of the first hop router to use when sending traffic to destinations outside its local subnet

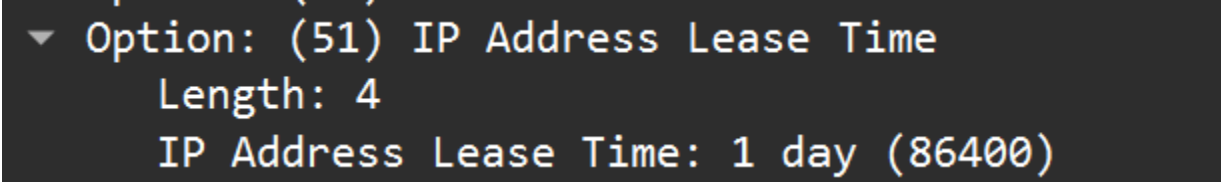
Without this, the client could only communicate with devices on its own local network and couldn't reach the internet

11. Yes, the client accepts the offered IP address!

The client sends back a DHCP Request message (not a decline)

Inside that Request, Option 50 (Requested IP Address) contains the same IP address that the server offered

This is the client's way of formally saying "yes, I want that address please"

12. 

IP Address Lease Time: 1 day (86400)

Purpose of lease time:

The lease time specifies how long the client is allowed to keep the assigned IP address. After it expires, the client must renew with the server or release the address back to the pool. This allows the server to recycle IP addresses from devices that have left the network.

13. Purpose of DHCP Release:

The DHCP Release message is sent by the client to tell the server it no longer needs the IP address, allowing the server to return it to the available pool.

Does the server acknowledge the release?

No, the DHCP server does NOT send an acknowledgment back to the client after receiving a Release message.

What if the Release message is lost?

Nothing immediately breaks, the client has already stopped using the address anyway. The server will simply hold that IP as "in use" until the lease time expires, at which point it automatically returns the address to the available pool.

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dhcpc-ethereal-trace-1.pcap

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Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.102	192.168.1.255	BROWSER	250	Domain/Workgroup Announcement WORKGROUP, NT Workstation, Domain...
2	7.587185	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x3e5e0ce3
3	7.588881	LinksysGroup_da:af:73	Broadcast	ARP	60	Who has 192.168.1.101? Tell 192.168.1.1
4	8.632950	192.168.1.1	255.255.255.255	DHCP	590	DHCP Offer - Transaction ID 0x3e5e0ce3
5	8.633123	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0x3e5e0ce3
6	8.635133	192.168.1.1	255.255.255.255	DHCP	590	DHCP ACK - Transaction ID 0x3e5e0ce3
7	8.638148	Dell_4f:36:23	Broadcast	ARP	42	ARP Announcement for 192.168.1.101
8	9.285757	Dell_4f:36:23	Broadcast	ARP	42	ARP Announcement for 192.168.1.101
9	10.285814	Dell_4f:36:23	Broadcast	ARP	42	ARP Announcement for 192.168.1.101
10	11.309600	192.168.1.101	224.0.0.22	IGMPv3	54	Membership Report / Join group 239.255.255.250 for any sources
11	11.311090	LinksysGroup_da:af:73	Broadcast	ARP	60	Who has 192.168.1.101? Tell 192.168.1.1
12	11.311102	Dell_4f:36:23	LinksysGroup_da:af:73	ARP	42	192.168.1.101 is at 00:08:74:4f:36:23
13	11.311569	192.168.1.1	192.168.1.101	ICMP	74	Destination unreachable (Protocol unreachable)
14	11.364272	192.168.1.101	192.168.1.255	NBNS	110	Registration NB NOHO<00>
15	11.895281	192.168.1.101	224.0.0.22	IGMPv3	54	Membership Report / Join group 239.255.255.250 for any sources
16	11.896474	192.168.1.1	192.168.1.101	ICMP	74	Destination unreachable (Protocol unreachable)
17	12.114052	192.168.1.101	192.168.1.255	NBNS	110	Registration NB NOHO<00>
18	12.864085	192.168.1.101	192.168.1.255	NBNS	110	Registration NB NOHO<00>
19	13.614118	192.168.1.101	192.168.1.255	NBNS	110	Registration NB NOHO<00>
20	14.364381	192.168.1.101	192.168.1.255	NBNS	110	Registration NB WORKGROUP<00>

Frame 3: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0

Ethernet II, Src: LinksysGroup_da:af:73 (00:06:25:da:af:73), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: LinksysGroup_da:af:73 (00:06:25:da:af:73)

Sender IP address: 192.168.1.1

Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)

Target IP address: 192.168.1.101

dhcpc-ethereal-trace-1.pcap Packets: 63 Profile: Default

14.

Yes, ARP packets are sent during the DHCP exchange.

Why does ARP appear?

After the server decides to offer an IP address to the client, it first sends an ARP request to check that the IP address it's about to offer is not already in use by another device on the network. This is called a gratuitous ARP or ARP probe.

Additionally, once the client receives its new IP address, it also sends an ARP request announcing its new IP to the network for two reasons:

- To check nobody else is already using that IP (conflict detection)
- To update the ARP caches of other devices on the network so they know the new IP-to-MAC mapping